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Constant reference tracking control for uncertain linear sampled-data systems

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ABSTRACT

This paper studies the reference-tracking control problem for uncertain sampled-data systems with partial observation of the states. We show that, by robustly stabilizing the closed-loop system, the proposed digital controller ensures that the controlled output tracks constant set points. Moreover, new design conditions given in terms of Differential Linear Matrix Inequalities are presented for obtaining controllers with partial and complete measurements of the states. The design conditions are illustrated by numerical examples.

1. Introduction

A significant problem in applied engineering is the reference tracking problem, in which a system output follows a constant set point. This is a traditional topic of research of control systems as illustrated by the large body of works for continuous-time systems, e.g., [1,2], and discrete-time systems [3,4], and the references therein. Some techniques employed for solving control goals involving tracking are using controllers with integral action [5], model-predictive control [1], linear quadratic regulators and/or reinforcement learning [6], the use of reference generators [7], a combination of the aforementioned approaches, among others. The problem of robust tracking control was also tackled in [8] for the continuous-time linear and nonlinear systems.

In the past decades, control systems have been usually implemented through digital devices, and due to the effects of sampling, classical solutions could lose effectiveness in this setting, [9,10]. For instance, the performance of controllers obtained through continuous-time techniques and subsequent discretization can be degraded [11]. Another usual approach is discretizing the system and directly employing a discrete-time framework for the control design. However, in this case, inter-sampling behavior is not considered which could be critical in certain applications, e.g., [12].

Considering reference tracking for sampled-data systems, in [13] an optimal reference tracking technique based on the minimization of the quadratic cost of the error signal is proposed, by considering the exact discretization of the plant. In [14] the tracking control with uncertain sampling time is considered by using two different approaches, namely, exact discretization and impulsive hybrid systems, similar as the one employed in [15,16]. However, the conditions in [14] require that the controller be provided so that the feasibility of some Linear Matrix Inequalities (LMIs) are then checked. In [17,18], the design of tracking controllers and degradation effects were carried out via the lifting technique, and similarly, in [19], the tracking problem was

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studied by a generalization of lifting. In [20,21], the tracking references problem is addressed as sampled-data PID-based control, while in [22] the authors employ a data-driven approach. Finally in, in [23], the tracking problem for non-linear networked control systems is considered with a focus on different protocols.

It is worth pointing out that the aforementioned works did not consider uncertainties in the system and study inter-sampling tracking behavior. This is one of the main challenges for sampled data systems in general as the system's trajectory depends non-linearly on the uncertain (polytopic) parameters [16,24]. To overcome this difficulty, a more recent trend in the literature is the use of Differential Linear Matrix Inequalities (DLMIs) as in [16,25,26] which provides solutions to robust controller design at the cost of computational complexity. By considering quadratic Lyapunov functions with time-independent matrices such as in [14], it is possible to restrict DLMIs to LMIs, which eases the computational cost at the expense of potentially more conservative results.

In this work, we consider the problem of designing robust controllers for uncertain sampled-data systems that ensure the asymptotic stability of the closed-loop system whilst guaranteeing constant reference tracking for outputs chosen by the designer. The main contributions are as follows.

- By employing a *discrete-time sequence* as the reference signal to the digital controller with integral action and partial state observations, we show that *the continuous-time tracked output signal* of the uncertain system converges at steady state to the desired set-point value, not only at the sampling instants, but in the whole interval between samples.
- We provide DLMI and LMI conditions for designing the robust controller, which can be easily solved by off-the-shelf software.

This work is organized as follows. We present some auxiliary facts about impulsive systems in Section 2. The problem formulation and main goals are discussed in Section 3. In Section 4, we present the analysis results for tracking control and conditions for designing robust controllers for the cases of partial and complete state observation. The illustrative example and numerical issues are discussed in Section 5. The paper is concluded in Section 6.

The notation used in this work is standard. For matrices and vectors, the operator $^\top$ denotes transposition, and for block matrices, the symmetric block is represented by $\star$. For square matrices $M \in \mathbb{R}^{n_m \times n_m}$, the symmetric sum is given by $\mathrm{Her}[M] = M + M^\top$. The spectral radius and spectrum of M are denoted by $\rho[M]$ and $\sigma[M]$, respectively. Block diagonal matrices are represented by $\mathrm{diag}(M, N)$. Positive (negative) definiteness conditions of symmetric matrices $P \in \mathbb{R}^{n_p \times n_p}$ are denoted by $P \succ 0$ ($P \prec 0$). The vector $e_i \in \mathbb{R}^{n_p}$ is the ith standard basis of $\mathbb{R}^{n_p}$. For any continuous-time function $f(t)$, we set $f[k] = f(t_k)$ for $k \in \mathbb{N}$. The standard simplex of dimension N is denoted by $\Lambda_N = \{\alpha = (\alpha_1, \dots, \alpha_N) : \sum_{i=1}^N \alpha_i = 1,\ \alpha_i \ge 0,\ i = 1, \dots, N\}$.Finally, for vectors $v(t,\alpha)$ depending on both the time $t \in \mathbb{R}^+$ and $\alpha \in \Lambda_N$, we omit the dependence on α to ease the notation; and for vectors $v(\alpha)$ depending only on $\alpha \in \Lambda_N$, we set $v_\alpha = v(\alpha)$.

2. Auxiliary results

In this section, we present some auxiliary results. For that, we introduce the following general uncertain periodic impulsive system

$$\begin{cases} \dot{\xi}(t) &= F(\alpha)\xi(t) \\ \xi(t_k) &= H(\alpha)\xi(t_k^-) + J(\alpha)r[k], \xi(t_0^-) = 0, \end{cases} \tag{1}$$

where $\xi(t) \in \mathbb{R}^{n_\xi}$ is the state and $r[k] \in \mathbb{R}^{n_r}$ is the input. The impulses occur at every $t = t_k$, $k \in \mathbb{N}$, with fixed fundamental period $t_{k+1} - t_k = h > 0$. Similarly as in [16,25], and [26], the uncertain matrix $F(\alpha)$ is modeled by

$$F(\alpha) = \sum_{i=1}^{N} \alpha_i F_i, \quad \alpha \in \Lambda_N. \tag{2}$$

Note that the state trajectory for every system realization depends on a time-invariant $\alpha \in \Lambda_N$ for all $t \in \mathbb{R}^+$. Therefore, we recall that we omit the dependence of $\xi(t)$ on α to ease the notation. The lemma below, adapted from [25], introduces conditions for verifying the asymptotic stability (a.s.) of (1).

Lemma 1. *For given $h > 0$ and $r \equiv 0$, if there exists a symmetric differentiable matrix function $P : [0,h) \to \mathbb{R}^{n_\xi \times n_\xi}$ that satisfies the following differential matrix inequalities*

$$F(\alpha)^\top P(t) + P(t)F(\alpha) + \dot{P}(t) \preceq 0 \tag{3}$$

in the interval $t \in [0,h)$, with boundary conditions

$$P(0) \succ 0,\ P(h) \succ H(\alpha)^\top P(0) H(\alpha), \tag{4}$$

for every $\alpha \in \Lambda_N$, then the system (1) is asymptotically stable.

As pointed out in [25], imposing $P(0) \succ 0$ (or $P(h) \succ 0$ as in [26]) ensures that $P(t) \succ 0$ for all $t \in [0,h)$. Moreover, the conditions of Lemma 1 become necessary for a.s. analysis in the case $N = 1$.

System (1) can be equivalently described at time instants $t = t_k^-$ by the discrete-time system, given by

$$\xi(t_{k+1}^-) = e^{F(\alpha)h} H(\alpha)\xi(t_k^-) + e^{F(\alpha)h} J(\alpha) r[k] \tag{5}$$

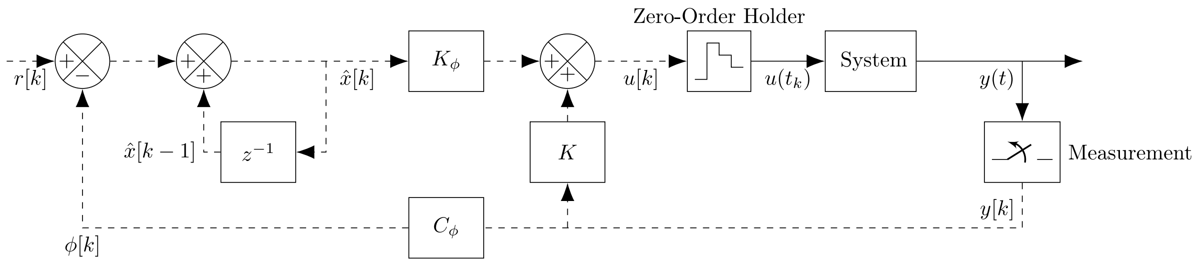

Fig. 1. Tracking-reference diagram for the sampled-data system. The dotted line represents a digital communication channel, while the solid line corresponds to continuous-time signals.

Therefore, the asymptotic stability of (1) is equivalent to $e^{F(\alpha)h}H(\alpha)$ being a Schur matrix. It follows that finding a solution to the conditions of Lemma 1 ensures that $\rho[e^{F(\alpha)h}H(\alpha)] < 1$ for every $\alpha \in \Lambda_N$. Bearing this discussion in mind, we introduce the following proposition.

Proposition 1. *Consider that $\rho[e^{F(\alpha)h}H(\alpha)] < 1$ for every $\alpha \in \Lambda_N$, and $r[k] = \bar{r}$, $k \geq 0$, where $\bar{r} \in \mathbb{R}^{n_r}$ is a known constant. We get that*

$$\underline{\xi_\alpha} \triangleq \lim_{k\to\infty} \xi(t_k^-) = \left(I - e^{F(\alpha)h}H(\alpha)\right)^{-1} e^{F(\alpha)h}J(\alpha)\bar{r} \tag{6}$$

for every $\alpha \in \Lambda_N$.

Proof. By evaluating $\xi(t_k^-)$ through (5) for $\xi_0^- = 0$ and $r[k] = \bar{r}$, we get for $k \geq 1$ that

$$\xi(t_k^-) = \left[\sum_{i=0}^{k-1} \left(e^{F(\alpha)h}H(\alpha)\right)^i\right] e^{F(\alpha)h}J(\alpha)\bar{r}.$$

Therefore, since $\rho\left[e^{F(\alpha)h}H(\alpha)\right] < 1$, $\alpha \in \Lambda_N$, the limit of $\xi(t_k^-)$ for $k \to \infty$ is well defined, as the related Neumann series in the last equation converge and, consequently, $\xi(t_k^-)$ converges to the constant value denoted by $\underline{\xi_\alpha}$. The exact value of $\underline{\xi_\alpha}$ shown in (6) can be calculated by either taking the limit of $k \to \infty$ on both sides of (5) and isolating $\underline{\xi_\alpha}$, or employing the explicit solution of the Neumann series, see [27]. □

Note that the values $\underline{\xi_\alpha}$ in (6) depend non-linearly on α. Moreover, considering (1) and (6), we get that

$$\bar{\xi}_\alpha \triangleq \lim_{k\to\infty} \xi(t_k) = H(\alpha)\underline{\xi_\alpha} + J(\alpha)\bar{r},\ \alpha \in \Lambda_N. \tag{7}$$

3. Problem statement

Let a linear time-invariant system be as in Fig. 1 with a uncertain time-invariant plant,

$$\begin{cases} \dot{x}(t) &= A(\alpha)x(t) + B(\alpha)u(t), \\ y(t) &= C_y(\alpha)x(t),\ x(0) = 0, \end{cases} \tag{8}$$

where $x(t) \in \mathbb{R}^{n_x}$ is the state vector, $u(t) \in \mathbb{R}^{n_u}$ is the control input, and $y(t) \in \mathbb{R}^{n_y}$ is the measured output. Moreover, $A(\alpha)$, $B(\alpha)$, and $C_y(\alpha)$ are uncertain and described similarly as $F(\alpha)$ in (2). We assume that $C_y(\alpha)$, has full row rank, $\alpha \in \Lambda_N$, and recall that we omit the dependence of any time-varying vector on α.

The control signal is a discrete-time sequence $u[k]$ generated through a digital device, so that

$$u(t) = u[k],\ \forall t \in [t_k, t_{k+1}),\ k \in \mathbb{N}, \tag{9}$$

where t_k is the sequence of sampling instants belonging to the set $\mathcal{T}$, such that $\mathcal{T} : \{t_k\}_{k\in\mathbb{N}}$ with consecutive sampling times respecting $t_{k+1} - t_k = h > 0$, for all $k \in \mathbb{N}$. Thus, the digital control structure is given by

$$\begin{cases} \hat{x}[k] &= \hat{x}[k-1] + r[k] - \phi[k] \\ u[k] &= K_\phi\hat{x}[k-1] + Ky[k],\ \hat{x}[0] = 0, \end{cases} \tag{10}$$

where $\hat{x}[k] \in \mathbb{R}^{n_\phi}$, and $r[k] \in \mathbb{R}^{n_\phi}$ is the set-point signal defined by,

$$r[k] = \begin{cases} \bar{r}, & k \geq 0 \\ 0, & \text{otherwise}, \end{cases} \tag{11}$$

for a fixed known $\bar{r} \in \mathbb{R}^{n_\phi}$; and the signal $\phi[k] = \phi(t_k)$, $\phi(t) \in \mathbb{R}^{n_\phi}$, is the regulated output, given by

$$\phi(t) = C_\phi y(t), \tag{12}$$

where matrix C_ϕ selects the components of $y(t)$ chosen by the designer to track the reference signals. That is, C_ϕ has the following structure,

$$C_\phi = \begin{bmatrix} e_{i_1} & \cdots & e_{i_{n_\phi}} \end{bmatrix}^\top, \; e_i \in \mathbb{R}^{n_y}, \tag{13}$$

where the indices $i_1, \ldots, i_{n_\phi} \in \{1, \ldots, n_y\}$ are the desired positions of $y(t)$; and e_i is the unitary vector of size n_y with one in the ith position, and zero elsewhere. Therefore, by connecting (8) to the sampled-data controller (9), defining the extended state $\xi(t) = \begin{bmatrix} x(t)^\top & u(t)^\top & \hat{x}(t)^\top \end{bmatrix}^\top \in \mathbb{R}^{n_\xi}$, with $n_\xi = n_x + n_u + n_\phi$, and noting that $\phi(t) = \phi[k]$, $t \in [t_k, t_{k+1})$, $k \in \mathbb{N}$, the closed-loop system can be written as an uncertain hybrid linear system given by,

$$\begin{cases} \dot{\xi}(t) & = F(\alpha)\xi(t) \\ \xi(t_k) & = H(\alpha)\xi(t_k^-) + Jr[k], \end{cases} \tag{14}$$

where, by defining $C_{\phi y}(\alpha) \triangleq C_\phi C_y(\alpha)$, we set

$$F(\alpha) = \begin{bmatrix} A(\alpha) & B(\alpha) & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, H(\alpha) = \begin{bmatrix} I & 0 & 0 \\ KC_y(\alpha) & 0 & K_\phi \\ -C_{\phi y}(\alpha) & 0 & I \end{bmatrix}, \tag{15}$$

and $J = \begin{bmatrix} 0 & 0 & I \end{bmatrix}^\top$. We say that the pair (K, K_ϕ) robustly stabilizes (14) if $\rho[e^{F(\alpha)h}H(\alpha)] < 1$, $\alpha \in \Lambda_N$.

We are now able to formally state the main goal of this work, that is, finding a controller as in (10) that robustly stabilizes the closed-loop system in (14), while also ensuring that the regulated output $\phi(t)$ tracks the constant set-points:

$$\lim_{t\to\infty} \phi(t) = \bar{r}. \tag{16}$$

By noting that the closed-loop system in (14) has the same form as (1) with $J(\alpha) = J$ for all $\alpha \in \Lambda_N$, a starting point for finding robust stabilizing controllers is considering K and K_ϕ as decision variables in (3)–(4). Such an approach would amount to solving a nonlinear matrix differential feasibility problem with an infinite number of constraints, since there are products involving (K, K_ϕ) and matrices $P(0)$, and (3)–(4) must be solved for every $\alpha \in \Lambda_N$. As we shall see in the next section, we present convex solutions in terms of DLMIs that ensure the stability of the closed-loop system and the tracking condition (16).

Remark 1. The controller in (10) can be viewed through the lens of the so-called *Internal Model Principle* (IMP) in the sense that it has embedded in its structure a model of the signal to be tracked. More specifically, the controller (10) implements the integral action $\hat{x}[k] = \hat{x}[k-1] + e[k]$, where $e[k] = r[k] - \phi[k]$, that is akin to the constant reference $r[k] = r[k-1]$, $r[0] = r$, in (11), for $e(k) = 0$. For a modern treatment of IMP, we refer the reader to [8].

4. Main results

This section presents the main result of tracking analysis and control. In Section 4.1, we present the analysis result concerning tracking control through the controller structure (10). Moreover, in Section 4.2, we present DLMI conditions to calculate the controller given in (10) for partial and complete observation of the states.

4.1. Analysis results for tracking control

This section shows that robust stability with the controller given as in (10) ensures that the tracking condition (16) is met. For that, we introduce the matrices

$$A_t(\alpha) \triangleq e^{A(\alpha)t}, \; B_t(\alpha) \triangleq \int_0^t e^{A(\alpha)\tau} B(\alpha) d\tau$$

for $t \in [0, h]$, along with $C_s(\alpha) \triangleq \begin{bmatrix} A(\alpha) - Is & B(\alpha) \end{bmatrix}$ and

$$\mathcal{D}_{t,z}(\alpha) \triangleq \begin{bmatrix} Iz - A_t(\alpha) & -B_t(\alpha) \end{bmatrix} \tag{17}$$

for $s, z \in \mathbb{C}$, $t \in [0, h]$. We have the following auxiliary result.

Proposition 2. *We get that $v_\alpha \in \ker[C_0(\alpha)]$ if and only if $v_\alpha \in \ker[\mathcal{D}_{t,1}(\alpha)]$, for all $t \in [0, h]$, $\alpha \in \Lambda_N$.*

Proof. First, consider the partition $v_\alpha = \begin{bmatrix} v_{x\alpha}^\top & v_{u\alpha}^\top \end{bmatrix}^\top$, with $v_{x\alpha} \in \mathbb{R}^{n_x}$ and $v_{u\alpha} \in \mathbb{R}^{n_u}$ and note, by straightforward manipulations, that

$$B_t(\alpha) = \int_0^t e^{A(\alpha)\tau} B(\alpha) d\tau = \sum_{i=0}^{\infty} \frac{t^{i+1}}{(i+1)!} A(\alpha)^i B(\alpha).$$

Note that

$$\mathcal{D}_{t,1}(\alpha) v_\alpha = (I - A_t(\alpha)) v_{x\alpha} - B_t(\alpha) v_{u\alpha}$$

$$= -\sum_{i=0}^{\infty} \frac{t^{i+1}}{(i+1)!} A(\alpha)^i C_0(\alpha) v_\alpha. \tag{18}$$

If $v_\alpha \in \ker[C_0(\alpha)]$, then $v_\alpha \in \ker[\mathcal{D}_{t,1}(\alpha)]$, $t \in [0,h]$. Conversely, if $v_\alpha \in \ker[\mathcal{D}_{t,1}(\alpha)]$, $t \in [0,h]$, it must be that $v_\alpha \in \ker[C_0(\alpha)]$. □

For the following proposition, which introduces an explicit formula for calculating $\left(I - e^{F(\alpha)h} H(\alpha)\right)^{-1}$ in (6) for the system matrices in (15), we define the permutation matrix

$$\mathcal{T} = \begin{bmatrix} I & 0 & 0 \\ 0 & 0 & I \\ 0 & I & 0 \end{bmatrix}.$$

Proposition 3. *If $\rho[e^{F(\alpha)h} H(\alpha)] < 1$, $\alpha \in \Lambda_N$, then*

$$\left(I - e^{F(\alpha)h} H(\alpha)\right)^{-1} = \mathcal{T} \begin{bmatrix} \left(I - \tilde{A}(\alpha)\right)^{-1} & 0 \\ \tilde{B}(\alpha)\left(I - \tilde{A}(\alpha)\right)^{-1} & I \end{bmatrix} \mathcal{T} \tag{19}$$

for $\alpha \in \Lambda_N$, where,

$$\tilde{A}(\alpha) \triangleq \begin{bmatrix} A_h(\alpha) + B_h(\alpha) K C_y(\alpha) & B_h(\alpha) K_\phi \\ -C_{\phi y}(\alpha) & I \end{bmatrix}, \tilde{B} \triangleq \begin{bmatrix} C_y(\alpha)^\top K^\top \\ K_\phi^\top \end{bmatrix}^\top. \tag{20}$$

Proof. Note that

$$\mathcal{T} e^{F(\alpha)h} H(\alpha) \mathcal{T} = \begin{bmatrix} \tilde{A}(\alpha) & 0 \\ \tilde{B}(\alpha) & 0 \end{bmatrix}, \tag{21}$$

Now, due to the fact the system (14) is asymptotically stable and $\mathcal{T}$ is a simple permutation matrix, we note that $\rho[e^{F(\alpha)h} H(\alpha)] < 1$. Moreover, due to the block triangular structure of (21), we get that the elements of the spectrum $\sigma[\tilde{A}(\alpha)]$ are also in $\sigma[e^{F(\alpha)h} H(\alpha)]$. Therefore, $\rho[e^{F(\alpha)h} H(\alpha)] < 1$ implies that $\rho[\tilde{A}(\alpha)] < 1$, $\alpha \in \Lambda_N$, so that $(I - \tilde{A}(\alpha))^{-1}$ is well defined. Thus, by employing standard block matrix inversion results, it follows that

$$\left(I - \mathcal{T} e^{F(\alpha)h} H(\alpha) \mathcal{T}\right)^{-1} = \begin{bmatrix} I - \tilde{A}(\alpha) & 0 \\ -\tilde{B}(\alpha) & I \end{bmatrix}^{-1} = \begin{bmatrix} \left(I - \tilde{A}(\alpha)\right)^{-1} & 0 \\ \tilde{B}(\alpha)\left(I - \tilde{A}(\alpha)\right)^{-1} & I \end{bmatrix}$$

By multiplying the last equation by $\mathcal{T}$ to the left- and right-hand side, noting that $\mathcal{T}^{-1} = \mathcal{T}$, so that $(I - \mathcal{T} e^{F(\alpha)h} H(\alpha) \mathcal{T})^{-1} = \mathcal{T}(I - e^{F(\alpha)h} H(\alpha))^{-1} \mathcal{T}$, we get 3, and the claim follows. □

From Proposition 3, we get that if $\rho[e^{F(\alpha)h} H(\alpha)] < 1$, then there exist matrices $(M_1, M_2, M_3, M_4)(\alpha)$ satisfying

$$\left(I - \tilde{A}(\alpha)\right)^{-1} = \begin{bmatrix} M_1(\alpha) & M_2(\alpha) \\ M_3(\alpha) & M_4(\alpha) \end{bmatrix}, \tag{22}$$

that is, for all $\alpha \in \Lambda_N$,

$$\begin{bmatrix} I - A_h(\alpha) - B_h(\alpha) K C_y(\alpha) & -B_h(\alpha) K_\phi \\ C_{\phi y}(\alpha) & 0 \end{bmatrix} \begin{bmatrix} M_1(\alpha) & M_2(\alpha) \\ M_3(\alpha) & M_4(\alpha) \end{bmatrix} = I. \tag{23}$$

In particular, from (23), it follows that,

$$C_{\phi y}(\alpha) M_1(\alpha) = 0,\ C_{\phi y}(\alpha) M_2(\alpha) = I \tag{24}$$

so that $M_1(\alpha)$ and $M_2(\alpha)$ are well-defined due to the full row rank structure of $C_y(\alpha)$ and C_ϕ in (13). We now introduce the intermediary result.

Lemma 2. *If $\rho[e^{F(\alpha)h} H(\alpha)] < 1$, $\alpha \in \Lambda_N$, then by applying the input (11) to (14), we get that,*

$$\underline{\xi}_\alpha = \bar{\xi}_\alpha, \quad \alpha \in \Lambda_N, \tag{25}$$

where $\underline{\xi}_\alpha$ and $\bar{\xi}_\alpha$ are respectively given in (6) and (7).

Proof. Set $\zeta(t) \triangleq \mathcal{T}\xi(t)$, $\tilde{H}(\alpha) = \mathcal{T} H(\alpha) \mathcal{T}$, and $\tilde{J} = \mathcal{T} J$. Similarly as $\underline{\xi}_\alpha$ in (6) and $\bar{\xi}_\alpha$ in (7), set $\underline{\zeta}_\alpha = \lim_{k\to\infty} \zeta(t_k^-)$ and $\bar{\zeta}_\alpha = \lim_{k\to\infty} \zeta(t_k)$. Since (14) is a.s. and $\mathcal{T} = \mathcal{T}^\top = \mathcal{T}^{-1}$, we get from (6) and (7) that

$$\begin{aligned} \underline{\zeta}_\alpha - \bar{\zeta}_\alpha &= \lim_{k\to\infty} \{[I - \tilde{H}(\alpha)]\zeta(t_k^-) - \tilde{J} r[k]\} \\ &= \{[I - \tilde{H}(\alpha)]\mathcal{T}(I - e^{F(\alpha)h} H(\alpha))^{-1} \mathcal{T}\mathcal{T} e^{F(\alpha)h} \mathcal{T} - I\} \tilde{J} \bar{r} \end{aligned} \tag{26}$$

where (26) follows from (6) and

$$I - \tilde{H}(\alpha) = \begin{bmatrix} 0 & 0 & 0 \\ C_{\phi y}(\alpha) & 0 & 0 \\ -KC_y(\alpha) & -K_\phi & I \end{bmatrix} = \begin{bmatrix} \tilde{H}_1(\alpha) & 0 \\ -\tilde{B}(\alpha) & I \end{bmatrix}, \quad \tilde{H}_1(\alpha) \triangleq \begin{bmatrix} 0 & 0 \\ C_{\phi y}(\alpha) & 0 \end{bmatrix}.$$

Thus, from Proposition 3, we get that

$$[I - \tilde{H}(\alpha)]\mathcal{T}(I - e^{F(\alpha)h}H(\alpha))^{-1}\mathcal{T} = \begin{bmatrix} \tilde{H}_1(\alpha)\big(I - \tilde{A}(\alpha)\big)^{-1} & 0 \\ 0 & I \end{bmatrix}$$

and more specifically, from (22)–(24), it follows that

$$\tilde{H}_1(\alpha)\big(I - \tilde{A}(\alpha)\big)^{-1} = \begin{bmatrix} 0 & 0 \\ C_{\phi y}(\alpha)M_1(\alpha) & C_{\phi y}(\alpha)M_2(\alpha) \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & I \end{bmatrix},$$

Moreover, we get that $\mathcal{T}^\top e^{F(\alpha)h}\mathcal{T} = e^{F(\alpha)h}$ and $\tilde{J} = \begin{bmatrix} 0 & I & 0 \end{bmatrix}^\top$, so that by considering the previous manipulations, we get that $\underline{\zeta}_\alpha - \bar{\zeta}_\alpha = 0$. After straightforward manipulations, we get (25). □

The following theorem shows that asymptotic stability of (14) and the controller structure in (10) ensures the tracking condition (11).

Theorem 1. *If there exists a sampling period $h > 0$ and a pair $(K,\ K_\phi)$ that robustly stabilizes (14), for $r[k]$ as in (11), we have that (i) $\xi(t) \to \underline{\xi}_\alpha$ for $t \to \infty$ and every $\alpha \in \Lambda_N$; and (ii) the tracking condition (16) holds true.*

Proof. Considering that there exist $h > 0$ and (K, K_ϕ) such that (14) is stable for all $\alpha \in \Lambda_N$, we integrate $\dot{\xi}(t)$ in the interval $t \in [t_k, t_{k+1})$, which yields $\xi(t) = e^{F(\alpha)(t-t_k)}\xi(t_k)$. By setting $t = t_{k+1}^-$, taking the limit of $k \to \infty$ and considering Lemma 2, we get that

$$(I - e^{F(\alpha)h})\underline{\xi}_\alpha = 0, \tag{27}$$

where $\underline{\xi}_\alpha$ is given by (6). Set $\underline{\xi}_\alpha = \begin{bmatrix} \underline{x}_\alpha{}^\top & \underline{u}_\alpha{}^\top & \underline{\hat{x}}_\alpha{}^\top \end{bmatrix}^\top$ and note that (27) is fulfilled for any $\underline{\hat{x}}_\alpha \in \mathbb{R}^{n_\phi}$, as well as $\underline{x}_\alpha \in \mathbb{R}^{n_x}$ and $\underline{u}_\alpha \in \mathbb{R}^{n_u}$ respecting

$$v_\alpha = \begin{bmatrix} \underline{x}_\alpha{}^\top & \underline{u}_\alpha{}^\top \end{bmatrix}^\top \in \ker[\mathcal{D}_{t,1}(\alpha)] \tag{28}$$

for $t = h$. The existence of $h > 0$ and the pair (K, K_ϕ) such that (14) is stable necessarily implies that (i) the pair $(A(\alpha), B(\alpha))$ is stabilizable and (ii) h is a non-pathological sampling period for all $\alpha \in \Lambda_N$, so that $(A_h(\alpha), B_h(\alpha))$ is also stabilizable (see [28]). By the PBH lemma (see [29, Corollary 2.15]), we have that $\mathrm{rank}[C_s(\alpha)] = n_x$ for all s in the (complex) closed right half-plane; and $\mathrm{rank}[\mathcal{D}_{h,z}(\alpha)] = n_x$ for all $z \in \mathbb{C}$, $|z| \geq 1$. It follows that $\mathrm{rank}[C_0(\alpha)] = \mathrm{rank}[\mathcal{D}_{h,1}(\alpha)] = n_x$ and thus the matrix series in (18) has full rank. Therefore, since (28) holds for $t = h$, we have that $v_\alpha \in \ker[C_0(\alpha)]$, so that from Proposition 2, (28) holds for all $t \in [0, h]$. Set $t = \tau + t_k$, $t \in [t_k, t_{k+1})$, $k \in \mathbb{N}$, where $\tau \in [0, h)$ is the elapsed time since the last jump. Clearly the convergence of $\xi(\tau, t_k) = e^{F(\alpha)\tau}\xi(t_k)$ to a constant value for all $\tau \in [0, h]$ as $k \to \infty$ implies that $\xi(t) = \xi(\tau, t_k)$ converges to the same value. From (28), we get that $(I - e^{F(\alpha)\tau})\underline{\xi}_\alpha = 0$ for all $\tau \in [0, h]$ so that $\xi(\tau, t_k) \to \underline{\xi}_\alpha$ as $k \to \infty$. Thus,

$$\lim_{t\to\infty} \xi(t) = \underline{\xi}_\alpha. \tag{29}$$

Moreover, in view of (10) and Lemma 2, it follows that

$$\begin{aligned} \lim_{k\to\infty} \phi[k] &= \lim_{k\to\infty} \{r[k] + \hat{x}[k-1] - \hat{x}[k]\} \\ &= \bar{r} + \underline{\hat{x}}_\alpha - \bar{\hat{x}}_\alpha = \bar{r}. \end{aligned} \tag{30}$$

By recalling the definition of $\phi(t)$ in (12), we note that $\phi_\alpha \triangleq \lim_{t\to\infty} \phi(t) = C_{\phi y}(\alpha)\underline{x}_\alpha$, which, in view of restriction (30), evaluates to $\phi_\alpha = \bar{r}$, $\alpha \in \Lambda_N$, so that (16) holds, and the claim follows. □

The results of Theorem 1 state that, if there exist $h > 0$ and (K, K_ϕ) that robustly stabilize the closed-loop system in (14), then the tracking condition (16) holds. Therefore, as we will see in Section 4.2, the design strategy is to search for solutions to DLMI conditions that, in turn, ensure the existence of the stabilizing tracking controller.

4.2. Robust tracking control

We present DLMI conditions for calculating the controller in (10) in the following theorem.

Theorem 2. *If there exists a differentiable positive definite matrix $W(t) : [0, h) \to \mathbb{R}^{n_\xi \times n_\xi}$, matrices $M \in \mathbb{R}^{n_m \times n_m}$, $U \in \mathbb{R}^{n_{y\phi} \times n_{y\phi}}$ and $V \in \mathbb{R}^{n_{y\phi} \times n_u}$, with $n_m = n_x + n_\phi$, $n_{y\phi} = n_y + n_\phi$,* *and a given scalar* ε *such that*

$$W(t)F(\alpha)^\top + F(\alpha)W(t) - \dot{W}(t) \preceq 0, \tag{31}$$

holds for all $t \in [0,h)$ *and* $\alpha \in \Lambda_N$, *with the LMI conditions*

$$\begin{bmatrix} W(h) & W(h)C_{x\phi}^\top \\ \star & M \end{bmatrix} \succ 0, \tag{32}$$

$$\begin{bmatrix} \Phi(\alpha) & \Pi_1(\alpha) + \varepsilon \Pi_2^\top \\ \star & -\varepsilon(U + U^\top) \end{bmatrix} \succ 0 \tag{33}$$

where

$$\Phi(\alpha) = \begin{bmatrix} M & M\breve{K}(\alpha)^\top + \begin{bmatrix} 0_{n_{y\phi}\times n_x} & \breve{C}_y(\alpha)^\top V & 0_{n_{y\phi}\times n_\phi} \end{bmatrix} \\ \star & W(0) \end{bmatrix}, \Pi_1(\alpha) = \begin{bmatrix} M\breve{C}_y(\alpha)^\top - \breve{C}_y(\alpha)^\top U \\ 0_{n_\xi \times n_{y\phi}} \end{bmatrix},$$

$$\Pi_2^\top = \begin{bmatrix} 0_{n_m\times n_{y\phi}} \\ \begin{bmatrix} 0_{n_x\times n_{y\phi}} \\ V^\top \\ 0_{n_\phi\times n_{y\phi}} \end{bmatrix} \end{bmatrix}, \ \breve{K}(\alpha) = \begin{bmatrix} I_{n_x} & 0 \\ 0 & 0 \\ -C_{\phi y}(\alpha) & I_{n_\phi} \end{bmatrix}, \ \breve{C}_y(\alpha) = \begin{bmatrix} C_y(\alpha) & 0 \\ 0 & I_{n_\phi} \end{bmatrix}$$

$$C_{x\phi} \triangleq \begin{bmatrix} I_{n_x} & 0_{n_x\times n_u} & 0_{n_x\times n_\phi} \\ 0_{n_\phi\times n_x} & 0_{n_\phi\times n_u} & I_{n_\phi} \end{bmatrix}. \tag{34}$$

Then, by setting $\mathcal{K}^\top = \begin{bmatrix} K & K_\phi \end{bmatrix}^\top = U^{-1}V$, *we get the controller in (10) that stabilizes (14) and ensures that the tracking condition (16) is fulfilled.*

Proof. Consider that (31)–(33) holds so that, from (33), $W(0) \succ 0$ implying that $W(t) \succ 0$ for all $t \in [0,h)$; and also that U is non-singular. Therefore, note that (33) can be equivalently rewritten as

$$\begin{bmatrix} \Phi(\alpha) & \Psi(\alpha)U + \varepsilon\Pi_2^\top \\ \star & -\varepsilon(U+U^\top) \end{bmatrix} \succeq 0, \Psi(\alpha) = \begin{bmatrix} M\breve{C}_y(\alpha)^\top U^{-1} - \breve{C}_y(\alpha)^\top \\ 0 \end{bmatrix}. \tag{35}$$

Multiply (35) to the left-hand side by $\begin{bmatrix} I & \varepsilon^{-1}\Psi(\alpha) \end{bmatrix}$ and to the right-hand side by its transpose in order to get $\Phi(\alpha) + \mathrm{Her}[\Psi(\alpha)\Pi_2] \succeq 0$. By noting that $(M\breve{C}_y(\alpha)^\top - \breve{C}_y(\alpha)^\top U)\mathcal{K}^\top + \breve{C}_y(\alpha)^\top V = M\breve{C}_y(\alpha)^\top\mathcal{K}^\top$ and evaluating the last inequality, we get that

$$\begin{bmatrix} M & M\bar{K}(\alpha)^\top \\ \star & W(0) \end{bmatrix} \succeq 0, \quad \bar{K}(\alpha)^\top \triangleq \begin{bmatrix} I & C_y(\alpha)^\top K^\top & -C_{\phi y}(\alpha)^\top \\ 0 & K_\phi^\top & I \end{bmatrix}, \tag{36}$$

holds. Now multiply (31) to the left-hand and right-hand side by $W(t)^{-1}$, set $P(t) = W(t)^{-1}$ and recall that $\dot{P}(t) = -W(t)^{-1}\dot{W}(t)W(t)^{-1}$, so that (3) holds. Furthermore, note that $M \succ 0$ is a non-singular matrix due to (36). By applying the congruence transformations $\mathrm{diag}(W(h), I)^{-1}$ and $\mathrm{diag}(M, I)^{-1}$ to (32) and (36) respectively, followed by Schur complements in the resulting inequalities, yields

$$P(h) \succ C_{xv}^\top M^{-1} C_{xv} \succeq (C_{x\phi}^\top \bar{K}(\alpha)^\top) P(0) (\bar{K}(\alpha) C_{x\phi}), \tag{37}$$

where $\bar{K}(\alpha)$ is the block matrix in (36). It readily follows that (37) is exactly (4) so that, by Lemma 1, the closed-loop system (14) is a.s. The claim follows from Theorem 1. □

We now consider the case of perfect observation of the states, that is, $C_y(\alpha) = I_{n_x}$, $\alpha \in \Lambda_N$, presented in the following corollary.

Corollary 1. *If there exists a differentiable positive definite matrix* $W(t) : [0,h) \to \mathbb{R}^{n_\xi \times n_\xi}$, *matrices* $M \in \mathbb{R}^{(n_x+n_v)\times(n_x+n_v)}$, *and* $Z \in \mathbb{R}^{n_u \times n_x + n_v}$, *such that the DLMI in (31) holds for all* $t \in [0,h)$ *and* $\alpha \in \Lambda_N$, *along with the LMI conditions (32) and*

$$\begin{bmatrix} M & \begin{bmatrix} M\begin{bmatrix} I \\ 0 \end{bmatrix} & Z^\top & M\begin{bmatrix} -C_\phi^\top \\ I \end{bmatrix} \end{bmatrix} \\ \star & W(0) \end{bmatrix} \succ 0, \tag{38}$$

where C_{xv} *is given in (34). Then, by setting* $\begin{bmatrix} K & K_\phi \end{bmatrix} = ZM^{-1}$, *we get the controller in (10) that stabilizes (14) and ensures that the tracking condition (16) is fulfilled.*

Proof. Assuming that (31)–(32), and (38) hold, we note that $M \succ 0$ is a non-singular matrix due to (38). Set $ZM^{-1} = \begin{bmatrix} K & K_\phi \end{bmatrix}$, we get that (36) holds. The claim follows from the proof of Theorem 2. □

Remark 2. The conditions of Corollary 1 are also necessary for the existence controllers as in (10) and $C_y = I$. We note that (4) can be factored as the right-hand side of (37). By defining $M^{-1} = \bar{K}^\top P(0)\bar{K} + I\epsilon$, for a small $\epsilon > 0$ such that $P(h) \succ C_{xz}^\top[\bar{K}^\top P(0)\bar{K} + I\epsilon]C_{xz}$ still holds, an performing the inverse steps of the proof of Corollary 1, the claim follows.

5. Illustrative examples

We present in Section 5.1, the numerical procedure involved in solving (31), and in Section 5.2, a time simulation and feasibility analysis of Theorem 2 and Corollary 1.

5.1. Numerical implementations

For solving the design conditions of Theorem 2 and Corollary 1, we need to deal with the terms involving $\alpha \in \Lambda_N$ and depending on the time t. we assume $W(t)$ to be a λ–degree polynomial symmetric differentiable matrix, that is,

$$W(t) = \sum_{i=0}^{\lambda} W_i t^i, \quad \dot{W}(t) = \sum_{i=1}^{\lambda} i W_i t^{i-1}, \tag{39}$$

for all $t \in [0, h)$. Therefore, in order to solve the decision problem composed by the DLMIs in (31) and LMIs in (32)–(33), we follow the procedure described in [30] which considers that $0 \le t \le h$ is an uncertain parameter. More specifically, by defining $\gamma_1 = t/h, \quad \gamma_2 = 1 - \gamma_1$, $(\gamma_1, \gamma_2) \in \Lambda_2$, as well as the set $\mathcal{D}[\lambda] = \{(a, b) \in \mathbb{N} \times \mathbb{N} \,:\, a + b = \lambda\}$, it can be shown by straightforward manipulations that $W(t)$ in (39) can be equivalently rewritten as

$$W(t) = \sum_{(a,b)\in\mathcal{D}[\lambda]} \gamma_1^a \gamma_2^b W_{(a,b)}, \tag{40}$$

see page 3 in [30], where matrices $W_{(a,b)}$ are linear combinations of W_i in (39). Note that a similar procedure can also be done with $\dot{W}(t)$ in (39). Finally, we resort to the strategy used in [16,25,26] by properly changing $F(\alpha)$ to F_i and $W(t)$ to $W_{(a,b)}$ (and also $\dot{W}(t)$ to its respective vertices) in (31) and solve the resulting LMIs for all $i = 1, \ldots, N$ and $(a, b) \in \mathcal{D}[\lambda]$, which implies, through convexity arguments, and also by bearing in mind the change of variables in (40), that (31) holds. This whole procedure is done automatically by the parser ROLMIP, see [31] so that each decision variable $W_{(a,b)}$ (and the ones describing $\dot{W}(t)$) are found via a convex decision problem [32]. As part of this work, a 5th-degree polynomial approximation for $W(t)$ was implemented. For determining the parameter ε in Theorem 2, a search was carried out in the points $\{1, \ldots, 9\} \times \{10^{-3}, \ldots, 10^{6}\}$. In terms of computational complexity, considering the aforementioned adopted numerical approach, we have to solve $(\lambda + 1)N + 2$ LMIs from Theorem 2 with $\lambda + 2$ symmetric matrix variables and two full matrix variables of different dimensions. For Corollary 1, the number of LMIs and symmetric variables is the same, whereas just one full matrix variable is needed. Solving LMIs is a semidefinite programming problem, whose asymptotic complexity can be found in [32], and the references therein.

5.2. Numerical example

To illustrate the effectiveness of the proposal, two numerical experiments are presented. The first experiment provides a direct comparison with a solution from the literature, while the second experiment investigates the feasibility of the design conditions for different randomly generated systems. Therefore, in the first experiment, we consider the uncertain system (8) adapted from [20] with matrices

$$A_1 = \begin{bmatrix} -0.123 & 0.0542 \\ 0.0077 & -0.0137 \end{bmatrix}, \; A_2 = \begin{bmatrix} -0.05421 & 0.12300 \\ 0.00906 & -0.01238 \end{bmatrix}, \; B_1 = \begin{bmatrix} 0.0369 \\ 0 \end{bmatrix}, \; B_2 = \begin{bmatrix} 0.0762 \\ 0 \end{bmatrix} \; C_{y1} = \begin{bmatrix} 0.9 & 0 \end{bmatrix}, \; C_{y1} = \begin{bmatrix} 1.1 & 0 \end{bmatrix}$$

where the aim is to regulate the first state with $h = 2\,\mathrm{s}$, therefore we set $C_\phi = \begin{bmatrix} 1 & 0 \end{bmatrix}$. In [20], the authors design a robust sampled-data PID controller with gains $k_p = 0.08$ and $k_i = 1.6$. By employing Theorem 2 we obtain a controller with gains $K = \begin{bmatrix} -5.447 \end{bmatrix}$ and $K_\phi = \begin{bmatrix} 1.535 \end{bmatrix}$, while for Corollary 1 the controller gains are $K = \begin{bmatrix} -2.922 & -1.431 \end{bmatrix}$ and $K_\phi = \begin{bmatrix} 0.6499 \end{bmatrix}$. Fig. 2 shows the temporal behavior of this system controlled by the designed controller while tracking a piecewise constant reference given by $r[k] = 2$ for $0 \le k < 400$ and $r[k] = 1.5$ for $k \ge 400$. The curves for all techniques were obtained for different values of α to illustrate that robust regulation is effectively achieved. To establish a comparison metric, we used the Mean Standard Deviation (MSD). This is calculated by determining the standard deviation of the tracking error for different values of α at each moment in time, and then averaging all the standard deviations.

The temporal response of the system converges to the reference signal within a finite time for all $\alpha \in \Lambda_N$ across various scenarios. Despite this convergence, the responses exhibit distinct transient behaviors that depend on the specific values of α and each particular strategy. It is noteworthy that the PID strategy in [20] leads to longer settling times as well as larger overshoots for some values of α compared to the controllers provided by Theorem 2 and Corollary 1, whose time-responses are faster and do not present overshoot. Moreover, a more detailed analysis of the temporal signals reveals that the PID-based control strategy achieves a MSD, calculated considering all curves generated by the different values of α, of 0.4938. In contrast, the controllers derived from Theorem 2 and Corollary 1 yield MSDs of 0.3165 and 0.3518, respectively. This MSD metric allows for an evaluation of the robustness of each controller concerning parametric uncertainties when $\alpha \in \Lambda_N$, indicating that the proposed controller demonstrates superior robustness compared to the alternative approaches.

As a second experiment, we generated 100 random uncertain systems of order $n_x = 2$ and $n_u = 1$ with $N = 2$ and solve Theorem 2 ($n_y = 1$) and Corollary 1 ($n_y = 2$). In Theorem 2, we employ a ε search in the interval $\{1, \ldots, 9\} \times \{10^{-3}, \ldots, 10^{3}\}$. We have found 68 solutions to the conditions of Theorem 2 and 100 solutions to Corollary 1. We note that the feasibility rate of Theorem 2 can be improved by performing a broader line search on ε. Therefore, we got up to a 100% success rate, which illustrates the effectiveness of our method.

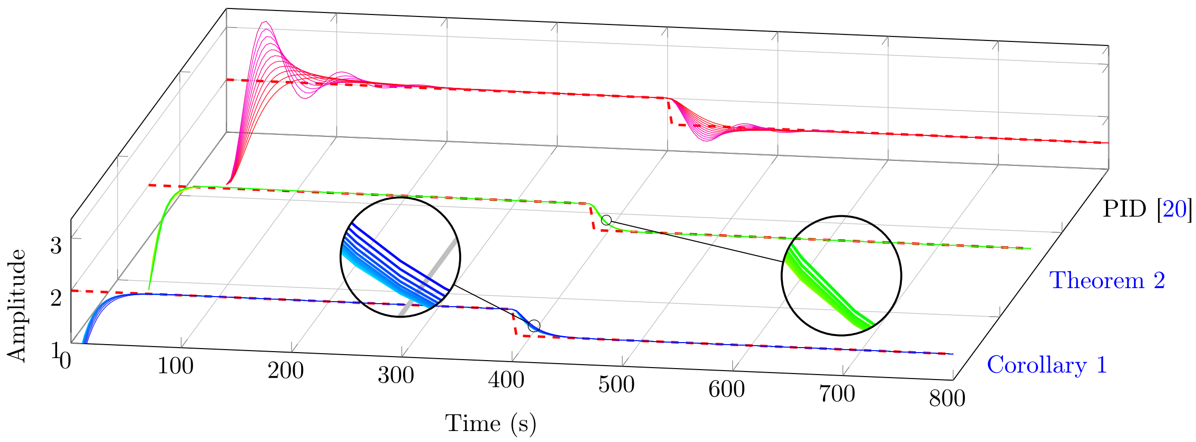

Fig. 2. Time response considering $C_\phi = \begin{bmatrix} 1 & 0 \end{bmatrix}$ and $h = 2$s. The dashed red line (- -) indicates the reference signal. The blue lines (—) correspond to the control when all state available for measurement for all $\alpha \in \Lambda_N$, the green lines (—) represent the trajectories when only the first state is available for all $\alpha \in \Lambda_N$, and the magenta lines (—) are the temporal response of the PID controller for all $\alpha \in \Lambda_N$.

6. Conclusions

In this work, we studied the tracking robust control by proposing a digital controller with integral action in the context of sampled-data systems. New design conditions for the proposed control structure based on DLMIs for full or partial state measurements are presented. The design conditions were tested by numerical examples. For future works, the problem of reference tracking for time-varying systems, inter-sampling performance analysis, a detailed analysis of the conditions that ensure the existence of stabilizing tracking controllers for the sampled-data setting, along with conditions that guarantee the existence of solutions to robust DLMIs, are topics of interest.

CRediT authorship contribution statement

Roberto M. Fuentes: Writing – original draft, Investigation, Writing – review & editing, Methodology. **Gabriela W. Gabriel:** Writing – review & editing, Validation, Investigation, Writing – original draft, Methodology. **André M. de Oliveira:** Writing – review & editing, Validation, Investigation, Writing – original draft, Methodology, Conceptualization. **Jonathan M. Palma:** Writing – review & editing, Supervision, Writing – original draft, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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